

# **Machine Learning for Oil and Gas Production Forecasting in the Permian Basin**

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## **Machine Learning for Oil and Gas Production Forecasting in the Permian Basin**

The National Energy Technology Laboratory (NETL) conducted a real-world machine-learning project to forecast daily oil, water, and natural-gas production from horizontal wells in the Midland Basin of West Texas. The study used a quasi-experimental design to compare models built with proprietary operator data against models using information more commonly available from public sources. Its central question was whether richer operational and geological data could materially improve production forecasting (Vikara et al., 2022).

### **Data Preprocessing**

The original dataset included 318 horizontal wells completed in the Spraberry/Dean and Wolfcamp formations between 2015 and 2020. After records with critical missing information were removed, 235 wells remained for model development. The data included production histories, pressure measurements, well-completion information, spatial variables, and well-log characteristics. Missing data were handled according to operational context. When a well was shut in, missing production values could reasonably be represented as zero. Missing flow measurements during active production were interpolated, while missing pressure values were estimated using multivariate imputation. Numerical variables were normalized to a 0–1 range. Because well histories differed in length, sequences were padded to a common length so they could be processed by the neural network (Vikara et al., 2022).

This preprocessing approach aligns with recommended practice because missing values were not treated with a single generic method. Instead, the researchers considered the physical meaning of the data. A limitation, however, is that the report does not describe a formal statistical method for detecting outliers. In future work, I would add an explicit outlier-screening process and

combine statistical detection with engineering review so that unusual but legitimate production events are not mistakenly removed.

### **Model Selection and Validation**

The researchers selected a Long Short-Term Memory (LSTM) neural network. This was appropriate because oil, water, and gas production are time-series variables, and LSTMs are designed to learn sequential relationships and changing patterns over time. Three model variants were created: a Proprietary Model, a Quasi-public Model, and a Public Model. The Proprietary Model included the richest operational, geological, pressure, and completion information, while the other models progressively reduced access to proprietary inputs. This allowed the researchers to isolate the value of additional data.

The wells were divided approximately 80% for training, 10% for validation, and 10% for holdout testing. Importantly, the split occurred at the well level, meaning the entire production history of a given well remained in only one partition. This strongly aligns with good validation practice because randomly splitting daily records from the same well could cause data leakage and artificially inflate test performance. Early stopping was also used to reduce overfitting (Vikara et al., 2022). One weakness is that model architecture and hyperparameters were refined largely through trial and error rather than systematic cross-validation or automated optimization. Although deep-learning models can be computationally expensive to tune, a more structured search process would strengthen confidence that the selected configuration was close to optimal.

### **Model Evaluation**

Because the study predicted continuous production volumes, regression metrics were more appropriate than classification measures such as accuracy, precision, recall, and F1-score. NETL

evaluated performance using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and  $R^2$  (coefficient of determination).

The Proprietary Model achieved the strongest overall results, with average  $R^2$  values of approximately 0.86 for oil and 0.79 for both water and gas production. It also produced substantially lower prediction errors than the Public Model. These findings indicate that high-frequency operational and geological information provided meaningful predictive value (Vikara et al., 2022). The study also showed that prediction quality weakened later in a well's producing life because fewer wells contained long production histories. This is an important reminder that strong average model performance can hide weaker performance in less-represented operating conditions.

### **Ethical Considerations**

The primary ethical issue in this project involved privacy and proprietary data governance. NETL obtained operator-specific information that was not publicly available, and the operator's identity was protected. Secure data-sharing arrangements were therefore necessary to balance research value with commercial confidentiality.

Traditional fairness concerns involving protected demographic groups are less applicable because the model predicts well production rather than human outcomes. However, model bias remains relevant. The data came from a specific operator and geological region, so the model may not perform equally well for other operators, formations, or basins. External validation would therefore be necessary before using the model elsewhere.

Environmental responsibility should also be considered because production forecasts can influence drilling, water management, and extraction decisions. ML outputs should support engineering judgment and regulatory requirements rather than replace them.

### **Post-Deployment Monitoring and Reflection**

The study does not provide a detailed post-deployment monitoring plan, which is its largest weakness relative to recommended ML practice. If deployed, I would continuously compare predicted and actual production using rolling RMSE and  $R^2$ . I would also monitor important input variables for data drift and track performance by formation, well age, and completion type. Predetermined error thresholds should trigger engineering review and possible retraining.

The practices I would apply in future work are context-aware missing-data treatment, well-level data separation, appropriate regression metrics, and early stopping. I would improve the approach by adding formal outlier detection, more systematic hyperparameter tuning, external validation, and a documented monitoring and retraining process. Overall, this case demonstrates that reliable machine learning depends not only on predictive accuracy but also on sound preprocessing, realistic validation, responsible data governance, and continued verification after deployment.

**AI Use Statement:** I used ChatGPT to help organize and refine this report. I verified the technical information against the original NETL source and completed the final analysis and reflection myself.

## References

Vikara, D., Bello, K., Wijaya, N., Warner, T., Sheriff, A., Hoffman, H., & Remson, D. (2022).

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